

# Probability & Stochastic Processes

## Final Exam Master Reference Guide

*Every Theorem You Need, Fully Worked*

### Scope of This Guide

Chapter 4: §4.2–4.3 (CDF/PDF), §4.6 (Gaussian RVs), §4.7 (Delta Functions & Mixed RVs)

Chapter 5: §5.7 ( $E[g(X,Y)]$ ), §5.8 (Covariance/Correlation), §5.9 (Bivariate Gaussian)

Chapter 6: §6.2 (Functions Yielding Continuous RVs), §6.3 (Functions Yielding Discrete/Mixed RVs)

Chapter 13: §13.7–13.11 (Stochastic Processes: Stationarity, WSS, Ergodicity, Cross-Correlation, Gaussian Processes)

### Professor's Exact Guidance (verbatim)

*“The final is made up. I will be giving you the exact sheet from the first exam, so anything from chapter 5, 6 and 13 — that’s what you should concentrate your notes on. There will be a section of the test where you will have a choice based on the material and chapters 4, 5 and 6. There will also be two questions on chapter 13.”*

Textbook: Yates & Goodman, *Probability and Stochastic Processes*, 3rd edition

Cross-checked against Exam 1 and Exam 2 solution keys

# Contents

|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| <b>How This Guide Is Organized</b>                                            | <b>1</b>  |
| <b>I Chapter 4 — Continuous Random Variables</b>                              | <b>2</b>  |
| 1 The CDF and PDF (§4.2–4.3): Foundations                                     | 2         |
| 1.1 The Cumulative Distribution Function . . . . .                            | 2         |
| 1.2 The Probability Density Function . . . . .                                | 2         |
| 1.3 Expected Value and Variance (§4.4) . . . . .                              | 3         |
| 1.4 Quick-Reference: Standard Continuous Families (§4.5) . . . . .            | 4         |
| 2 Gaussian Random Variables (§4.6)                                            | 5         |
| 3 Delta Functions & Mixed Random Variables (§4.7)                             | 8         |
| <b>II Chapter 5 — Multiple Random Variables</b>                               | <b>10</b> |
| 4 Expected Value of a Function of Two Random Variables (§5.7)                 | 10        |
| 5 Covariance, Correlation & Independence (§5.8)                               | 12        |
| 6 Bivariate Gaussian Random Variables (§5.9)                                  | 15        |
| <b>III Chapter 6 — Probability Models of Derived Random Variables</b>         | <b>17</b> |
| 7 Functions Yielding Continuous Random Variables (§6.2)                       | 17        |
| 8 Functions Yielding Discrete or Mixed Random Variables (§6.3)                | 20        |
| <b>IV Chapter 13 — Stochastic (Random) Processes</b>                          | <b>21</b> |
| 9 Setup: Mean and Autocorrelation of a Process (§13.7)                        | 21        |
| 10 Stationary Processes (§13.8)                                               | 22        |
| 11 Wide Sense Stationary Processes & Ergodicity (§13.9)                       | 24        |
| 11.1 Ergodicity and Time Averages . . . . .                                   | 25        |
| 12 Cross-Correlation (§13.10)                                                 | 28        |
| 13 Gaussian Processes (§13.11)                                                | 29        |
| 13.1 Master Comparison Table: Which Processes Are Stationary / WSS? . . . . . | 30        |
| <b>V Full Exam &amp; Homework Problem Bank — Every Problem, Fully Worked</b>  | <b>31</b> |
| 14 Exam 1 — Full Solutions (7 Problems)                                       | 31        |

|                                                             |               |
|-------------------------------------------------------------|---------------|
| <b>15 Exam 2 — Full Solutions (5 Problems)</b>              | <b>34</b>     |
| <b>16 Chapter 13 Homework — Full Solutions (6 Problems)</b> | <b>36</b>     |
| <br><b>VI Formula Cheat Sheet</b>                           | <br><b>40</b> |
| <b>Master Formula Reference</b>                             | <b>40</b>     |
| <b>Rapid Self-Check Questions</b>                           | <b>42</b>     |
| <b>Final Reminders</b>                                      | <b>43</b>     |

## How This Guide Is Organized

Every **Definition** appears in a **blue** rule-box, every **Theorem** in a **green** rule-box, every worked **Example** in a **purple** rule-box, and every explicit professor-emphasized remark in an **orange** rule-box. Theorems the professor personally flagged on the review sheet (Theorem 13.10, 13.11, 13.12, and Example 13.22) get a special double-rule **red** box with a ★ — **these are the highest-priority items to memorize**. Every Theorem/Definition number matches the exact number used in Yates & Goodman, 3rd edition, so you can cross-reference directly with your textbook or the professor's exam-1 sheet. Part **V** reproduces *every single problem* from Exam 1 and Exam 2 with complete, checked, step-by-step solutions, tagged by topic. Part **VI** is a dense one-page-style formula cheat-sheet for last-minute review.

## Part I

# Chapter 4 — Continuous Random Variables

## 1 The CDF and PDF (§4.2–4.3): Foundations

### 1.1 The Cumulative Distribution Function

---

#### Definition 4.1 — Cumulative Distribution Function (CDF)

The CDF of random variable  $X$  is

$$F_X(x) = P[X \leq x].$$

This is a probability model for *any* random variable — discrete, continuous, or mixed.  $F_X(x)$  is a continuous function *if and only if*  $X$  is a continuous random variable.

---



---

#### Theorem 4.1 — Basic CDF Properties

For any random variable  $X$ :

- (a)  $F_X(-\infty) = 0$
- (b)  $F_X(\infty) = 1$
- (c)  $P[x_1 < X \leq x_2] = F_X(x_2) - F_X(x_1)$ .

$F_X$  is nondecreasing; for *discrete*  $X$  it is a jump-staircase (jump size =  $P_X(x)$  at each  $x \in S_X$ ); for *continuous*  $X$  it is continuous everywhere (this continuity *is* the definition of “continuous r.v.”).

---



---

#### Definition 4.2 — Continuous Random Variable

$X$  is continuous if and only if its CDF  $F_X(x)$  is a continuous function of  $x$ .

---

### 1.2 The Probability Density Function

---

#### Definition 4.3 — Probability Density Function (PDF)

For a continuous random variable  $X$ ,

$$f_X(x) = \frac{dF_X(x)}{dx}.$$

The PDF is the single most useful complete probability model of a continuous r.v.: its graph shows directly which values are likely.

---



---

### Theorem 4.2 — Basic PDF Properties

- (a)  $f_X(x) \geq 0$  for all  $x$
  - (b)  $F_X(x) = \int_{-\infty}^x f_X(u) du$
  - (c)  $\int_{-\infty}^{\infty} f_X(x) dx = 1.$
- 
- 

### Theorem 4.3 — Interval Probability from the PDF

$$P[x_1 < X \leq x_2] = \int_{x_1}^{x_2} f_X(x) dx.$$


---



---

### Critical Exam Reminder

The PDF value  $f_X(x)$  is *not* a probability — it can exceed 1. Only *areas* under  $f_X$  are probabilities. Because  $P[X = x] = 0$  for continuous  $X$ , all four events  $\{x_1 < X < x_2\}$ ,  $\{x_1 \leq X < x_2\}$ ,  $\{x_1 < X \leq x_2\}$ ,  $\{x_1 \leq X \leq x_2\}$  have the *same* probability. This is **false** for discrete/mixed random variables — watch strict vs. non-strict inequalities there.

---

## 1.3 Expected Value and Variance (§4.4)

---

### Definition 4.4 — Expected Value

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

**Theorem 4.4 — Law of the Unconscious Statistician**

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

This lets you find  $\mathbb{E}[g(X)]$  *without* first deriving the PDF of  $g(X)$  — the single most time-saving trick on the exam.

**Theorem 4.5 — Linearity and Variance**

For any random variable  $X$ :

- |                                                 |                                                |
|-------------------------------------------------|------------------------------------------------|
| (a) $\mathbb{E}[X - \mu_X] = 0$                 | (b) $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$  |
| (c) $\text{Var}[X] = \mathbb{E}[X^2] - \mu_X^2$ | (d) $\text{Var}[aX + b] = a^2 \text{Var}[X]$ . |

**Key takeaway:** Always compute variance as  $\mathbb{E}[X^2] - (\mathbb{E}[X])^2$  — this is faster than integrating  $(x - \mu_X)^2 f_X(x)$  directly, and it is exactly how every exam solution does it.

**1.4 Quick-Reference: Standard Continuous Families (§4.5)**

| Family                   | PDF $f_X(x)$                                                | CDF $F_X(x)$                                                   | $\mathbb{E}[X]$     | $\text{Var}[X]$       |
|--------------------------|-------------------------------------------------------------|----------------------------------------------------------------|---------------------|-----------------------|
| Uniform( $a, b$ )        | $\frac{1}{b-a}, a < x < b$                                  | $\frac{x-a}{b-a}, a \leq x < b$                                | $\frac{a+b}{2}$     | $\frac{(b-a)^2}{12}$  |
| Exponential( $\lambda$ ) | $\lambda e^{-\lambda x}, x \geq 0$                          | $1 - e^{-\lambda x}, x \geq 0$                                 | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |
| Erlang( $n, \lambda$ )   | $\frac{\lambda^n x^{n-1} e^{-\lambda x}}{(n-1)!}, x \geq 0$ | $1 - \sum_{k=0}^{n-1} \frac{(\lambda x)^k e^{-\lambda x}}{k!}$ | $\frac{n}{\lambda}$ | $\frac{n}{\lambda^2}$ |

**Useful bridges:** If  $X$  is uniform( $a, b$ ) with integers  $a, b$ , then  $K = \lceil X \rceil$  is discrete-uniform( $a+1, b$ ) (Thm 4.7). If  $X$  is exponential( $\lambda$ ), then  $K = \lceil X \rceil$  is geometric( $p$ ) with  $p = 1 - e^{-\lambda}$  (Thm 4.9). The CDF of an Erlang( $n, \lambda$ ) relates to a Poisson( $\lambda x$ ):  $F_X(x) = P[\text{Poisson}(\lambda x) \geq n]$  (Thm 4.11) — Geometric/Pascal (discrete-time) and Exponential/Erlang (continuous-time) both describe the same underlying Poisson arrival process (formalized in §13.4).

## 2 Gaussian Random Variables (§4.6)

### Why This Section Matters

The Gaussian family appears in more practical applications than any other. Its graph is the familiar bell curve. **This is one of the professor's named topics — expect a direct question.**

### Definition 4.8 — Gaussian Random Variable

$X$  is Gaussian( $\mu, \sigma$ ), written  $X \sim N[\mu, \sigma^2]$ , if

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}, \quad \mu \in \mathbb{R}, \sigma > 0.$$

Peak height  $1/(\sigma\sqrt{2\pi})$  occurs at  $x = \mu$ . Small  $\sigma \Rightarrow$  narrow, tall peak; large  $\sigma \Rightarrow$  wide, flat peak.

### Theorem 4.12 — Mean and Variance

If  $X$  is Gaussian( $\mu, \sigma$ ):  $E[X] = \mu$ ,  $\text{Var}[X] = \sigma^2$ .

### Theorem 4.13 — Linear Transformation of a Gaussian

If  $X$  is Gaussian( $\mu, \sigma$ ), then  $Y = aX + b$  is Gaussian( $a\mu + b, |a|\sigma$ ). Any linear transform of a Gaussian r.v. is again Gaussian — this is why one standard-normal table serves every Gaussian( $\mu, \sigma$ ).

### Definition 4.9 — Standard Normal Random Variable

$Z$  is the Gaussian(0, 1) random variable, so  $E[Z] = 0$ ,  $\text{Var}[Z] = 1$ .

**Definition 4.10 — Standard Normal CDF**

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2} du.$$

**Theorem 4.14 — Gaussian CDF via  $\Phi$** 

If  $X$  is Gaussian( $\mu, \sigma$ ), then

$$F_X(x) = \Phi\left(\frac{x - \mu}{\sigma}\right), \quad P[a < X \leq b] = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right).$$

The transform  $z = (x - \mu)/\sigma$  is  $x$  expressed as a number of standard deviations from  $\mu$ .

**Theorem 4.15 — Symmetry of  $\Phi$** 

$$\Phi(-z) = 1 - \Phi(z).$$

**Definition 4.11 — Complementary CDF ( $Q$ -function)**

$$Q(z) = P[Z > z] = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} e^{-u^2/2} du = 1 - \Phi(z).$$

**Key takeaway:**  $Q(z)$  (the tail) is where all the interesting exam action is: bit-error rates, outage probabilities, and “rare event” questions are always phrased in terms of  $Q$ , not  $\Phi$ . A “3-sigma event” has probability  $\approx 10^{-3}$ ; a “5-sigma event”  $\approx 10^{-7}$ . Memorize  $\Phi(1) = 0.8413$ ,  $\Phi(2) = 0.97725$ , and that  $2\Phi(1) - 1 \approx 0.683$  (the 68–95–99.7 rule).

**Example 4.15–4.18 —  $z$ -scores, tail probabilities, and inverting a bit-error spec**

(4.15) Test score  $x = 46 \sim \text{Gaussian}(61, 10)$ .  $z = (46 - 61)/10 = -1.5$ .

(4.16)  $P[X < 46] = \Phi(-1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.067$  (lowest 6.7%).

(4.17)  $P[51 < X < 71]$ : with  $Z = (X - 61)/10$ , this is  $P[-1 < Z < 1] = 2\Phi(1) - 1 = 0.683$ .

(4.18) Bit-error probability  $Q(\sqrt{\gamma/2}) \leq 10^{-6}$ . From the  $Q$ -table,  $Q(z) < 10^{-6}$  once  $z > 4.75$ . Require  $\sqrt{\gamma/2} \geq 4.75 \Rightarrow \gamma \geq 2(4.75)^2 = 45.125$ , i.e.  $\boxed{\gamma > 45}$ .



### 3 Delta Functions & Mixed Random Variables (§4.7)

#### Why This Section Matters

Delta functions let *every* formula in this course (stated with integrals) work uniformly for discrete, continuous, *and* mixed random variables. Exam 1's Problem 1 is built entirely on this section — expect something similar.

#### Definition 4.12 — Unit Impulse (Delta) Function

$$d_\epsilon(x) = \begin{cases} 1/\epsilon & -\epsilon/2 < x < \epsilon/2 \\ 0 & \text{otherwise} \end{cases}, \quad \delta(x) = \lim_{\epsilon \rightarrow 0} d_\epsilon(x).$$

By convention  $\int_{-\infty}^{\infty} \delta(x) dx = 1$ .

#### Theorem 4.16 — Sifting Property

For any continuous function  $g(x)$ :

$$\int_{-\infty}^{\infty} g(x) \delta(x - x_0) dx = g(x_0).$$

#### Definition 4.13 — Unit Step Function

$$u(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}.$$

#### Theorem 4.17 — Step/Delta Relationship

$$\int_{-\infty}^x \delta(v) dv = u(x), \quad \text{equivalently} \quad \delta(x) = \frac{du(x)}{dx}.$$

**Key takeaway:** For a *discrete* RV, translate the PMF into impulse language:  $F_X(x) = \sum_{x_i} P_X(x_i) u(x - x_i)$  and  $f_X(x) = \sum_{x_i} P_X(x_i) \delta(x - x_i)$ . Then *every* continuous-RV formula (expectation, etc.) applies unchanged and recovers the discrete formula via sifting.

### Theorem 4.18 — Equivalent Statements About a Probability Mass

For a random variable  $X$ , these are equivalent:

$$(a) P[X = x_0] = q \quad (b) P_X(x_0) = q \quad (c) F_X(x_0^+) - F_X(x_0^-) = q \quad (d) f_X(x_0) = q \delta(0).$$

### Definition 4.14 — Mixed Random Variable

$X$  is mixed iff  $f_X(x)$  contains *both* impulses and nonzero finite values.

### Example 4.21 — Mixed RV — phone call duration

1/3 of calls never connect (duration 0); otherwise (prob. 2/3) duration is uniform(0, 3) minutes. Let  $A$  = “answered,”  $P[A] = 2/3$ . By total probability, for  $0 \leq y < 3$ :

$$F_Y(y) = P[Y \leq y | A^c] P[A^c] + P[Y \leq y | A] P[A] = (1)\left(\frac{1}{3}\right) + \left(\frac{y}{3}\right)\left(\frac{2}{3}\right) = \frac{1}{3} + \frac{2y}{9}.$$

So  $f_Y(y) = \frac{1}{3}\delta(y) + \frac{2}{9}$  on  $[0, 3)$  (impulse = point mass at  $y = 0$ ; constant  $2/9$  = continuous density).  
Then

$$E[Y] = \frac{1}{3}(0) + \int_0^3 y \cdot \frac{2}{9} dy = 1 \text{ minute}.$$

### End-of-Section Checklist

For any random variable  $X$ :  $F_X(x) = P[X \leq x]$  always exists. If  $F_X$  is a **jump-staircase**  $\Rightarrow X$  is **discrete**. If  $F_X$  is **continuous everywhere**  $\Rightarrow X$  is **continuous**. If  $F_X$  is **piecewise continuous with some jumps**  $\Rightarrow X$  is **mixed** (its PDF then contains one or more impulses).

## Part II

## Chapter 5 — Multiple Random Variables

## 4 Expected Value of a Function of Two Random Variables (§5.7)

**Theorem 5.9 — Expected Value of  $W = g(X, Y)$** 

$$\text{Discrete: } E[W] = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) P_{X,Y}(x, y),$$

$$\text{Continuous: } E[W] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy.$$

**Theorem 5.10 — Linearity of Expectation (Two Variables)**

$$E[a_1 g_1(X, Y) + \cdots + a_n g_n(X, Y)] = a_1 E[g_1(X, Y)] + \cdots + a_n E[g_n(X, Y)].$$

**Theorem 5.11 — Expected Value of a Sum**

$$E[X + Y] = E[X] + E[Y].$$

This holds *even if  $X, Y$  are dependent*, and needs only the *marginal* models  $f_X, f_Y$  — not the joint model.

**Theorem 5.12 — Variance of a Sum**

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2 E[(X - \mu_X)(Y - \mu_Y)].$$

*Proof sketch.*  $\text{Var}[X + Y] = E[(X - \mu_X) + (Y - \mu_Y)]^2$ ; expand the square and apply Thm 5.10 term-by-term. The cross term  $E[(X - \mu_X)(Y - \mu_Y)]$  is the **covariance** — see §5.  $\square$

---

**Contrast to Memorize**

$E[X + Y]$  needs *only* the marginals (Thm 5.11) — but  $\text{Var}[X + Y]$  needs the *full joint* model, through the covariance term (Thm 5.12). This asymmetry is a favorite exam trap.

---

---

**Example 5.15 — Transmission time  $g(R, L)$ , and why  $E[g(X, Y)] \neq g(E[X], E[Y])$** 

Speed  $R$  (Mb/s):  $P[R = 5] = 0.4$ ,  $P[R = 10] = 0.6$ . Length  $L$  (kb), independent of  $R$ :  $P[L = 750] = 0.3$ ,  $P[L = 1500] = 0.5$ ,  $P[L = 2500] = 0.2$ . Transmission time is  $T = 8L/(1000R)$  seconds. Since  $R, L$  are independent,

$$E[T] = \frac{8}{1000} \left( \frac{0.4}{5} + \frac{0.6}{10} \right) (0.3(750) + 0.5(1500) + 0.2(2500)) = 1.652 \text{ s.}$$

But  $E[R] = 8$  Mb/s,  $E[L] = 1475$  kb, so  $g(E[R], E[L]) = 8(1475)/(1000 \cdot 8) = 1.475 \text{ s} \neq E[T] = 1.652 \text{ s}$ . **In general**  $E[g(X, Y)] \neq g(E[X], E[Y])$ .

---

## 5 Covariance, Correlation & Independence (§5.8)

---

### Definition 5.5 — Covariance

$$\text{Cov}[X, Y] = \text{E}[(X - \mu_X)(Y - \mu_Y)].$$

Positive  $\Rightarrow X, Y$  tend to move together; negative  $\Rightarrow$  opposite;  $\approx 0 \Rightarrow$  little linear relationship. *But* the raw magnitude of  $\text{Cov}[X, Y]$  depends on the measurement units of  $X, Y$  — motivating the correlation coefficient below.

---

### Definition 5.6 — Correlation Coefficient

$$\rho_{X,Y} = \frac{\text{Cov}[X, Y]}{\sqrt{\text{Var}[X]\text{Var}[Y]}} = \frac{\text{Cov}[X, Y]}{\sigma_X \sigma_Y}.$$

Dimensionless, unaffected by unit changes.

---

### Theorem 5.13 — Correlation Coefficient Under Linear Rescaling

If  $\hat{X} = aX + b$ ,  $\hat{Y} = cY + d$ :  $\rho_{\hat{X}, \hat{Y}} = \rho_{X,Y}$ ,  $\text{Cov}[\hat{X}, \hat{Y}] = ac \text{Cov}[X, Y]$ .

---

### Theorem 5.14 — Correlation Coefficient Is Bounded

$$-1 \leq \rho_{X,Y} \leq 1.$$


---

*Proof.* Let  $W = X - aY$ . Then  $\text{Var}[W] = \text{Var}[X] - 2a \text{Cov}[X, Y] + a^2 \text{Var}[Y] \geq 0$  for every  $a$ . Choosing  $a = \sigma_X/\sigma_Y$  gives  $\text{Cov}[X, Y] \leq \sigma_X \sigma_Y \Rightarrow \rho_{X,Y} \leq 1$ ; choosing  $a = -\sigma_X/\sigma_Y$  gives  $\rho_{X,Y} \geq -1$ .  $\square$

---

### Theorem 5.15 — Correlation of a Perfect Linear Relationship

If  $Y = aX + b$ :  $\rho_{X,Y} = -1$  (if  $a < 0$ ),  $0$  (if  $a = 0$ ),  $1$  (if  $a > 0$ ).

**Definition 5.7 — Correlation**

$$r_{X,Y} = E[XY].$$

**Theorem 5.16 — The Master Covariance/Variance Identities**

- (a)  $\text{Cov}[X, Y] = r_{X,Y} - \mu_X \mu_Y$
- (b)  $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y]$
- (c) If  $X = Y$  :  $\text{Cov}[X, Y] = \text{Var}[X] = \text{Var}[Y]$ ,  $r_{X,Y} = E[X^2]$ .

**Key takeaway:** Formula (a) above —  $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$  — is how *every* exam solution actually computes covariance in practice. Never integrate  $(x - \mu_X)(y - \mu_Y)$  directly; find  $E[XY]$ ,  $E[X]$ ,  $E[Y]$  separately and combine.

**Definition 5.8 / 5.9 — Orthogonal vs. Uncorrelated**

$X, Y$  are **orthogonal** if  $r_{X,Y} = 0$ .  $X, Y$  are **uncorrelated** if  $\text{Cov}[X, Y] = 0$ . (Confusingly named: orthogonal = zero *correlation*; uncorrelated = zero *covariance*.)

★ **Theorem 5.17 — Independence  $\Rightarrow$  Uncorrelated (Not Conversely) ★** [PROFESSOR-FLAGGED — MEMORIZE]

For independent random variables  $X, Y$ :

- (a)  $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$ ,      (b)  $r_{X,Y} = E[X]E[Y]$ ,
- (c)  $\text{Cov}[X, Y] = \rho_{X,Y} = 0$ ,      (d)  $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$ .

**Independent  $\Rightarrow$  uncorrelated, but  $\text{Cov}[X, Y] = 0$  does NOT imply independence in general.** (Exception: bivariate Gaussian — see Theorem 5.20.)

**Example 5.18 — Signal + noise:  $Y = X + Z$** 

$X \sim \text{Gaussian}(0, \sigma_X)$  (signal),  $Z \sim \text{Gaussian}(0, \sigma_Z)$  (noise), independent. By Thm 5.17(c),  $\text{Cov}[X, Z] = \rho_{X,Z} = 0$ . By Thm 5.17(d),  $\text{Var}[Y] = \sigma_X^2 + \sigma_Z^2$ . Then

$$\text{Cov}[X, Y] = \text{E}[X(X + Z)] = \text{E}[X^2] = \sigma_X^2, \quad \rho_{X,Y} = \sqrt{\frac{\sigma_X^2/\sigma_Z^2}{1 + \sigma_X^2/\sigma_Z^2}}.$$

**Interpretation:**  $\rho_{X,Y}$  is driven by the signal-to-noise ratio  $\sigma_X^2/\sigma_Z^2$  — large SNR  $\Rightarrow \rho_{X,Y} \rightarrow 1$  (observing  $Y$  tells you a lot about  $X$ ).

## 6 Bivariate Gaussian Random Variables (§5.9)

---



---

★ **Theorem 5.10 (Def.) — The Bivariate Gaussian PDF — KNOW THIS FORMULA COLD ★ [PROFESSOR-FLAGGED — MEMORIZE]**

$X, Y$  have a bivariate Gaussian PDF with parameters  $\mu_X, \mu_Y, \sigma_X > 0, \sigma_Y > 0$ , and  $-1 < \rho_{X,Y} < 1$  if

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{X,Y}^2}} \exp \left[ -\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - \frac{2\rho_{X,Y}(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2(1-\rho_{X,Y}^2)} \right].$$

This is the exact formula referenced on the professor's review sheet. Five parameters:  $\mu_X, \mu_Y, \sigma_X, \sigma_Y, \rho_{X,Y}$  — and  $\rho_{X,Y}$  really *is* the correlation coefficient of  $X, Y$  (Theorem 5.19).

---



---



---

### Theorem 5.18 — Marginals of a Bivariate Gaussian Are Gaussian

If  $X, Y$  are bivariate Gaussian (Def. 5.10), then  $X \sim \text{Gaussian}(\mu_X, \sigma_X)$  and  $Y \sim \text{Gaussian}(\mu_Y, \sigma_Y)$  — i.e. each marginal “forgets”  $\rho_{X,Y}$  entirely.

---



---



---

### Theorem 5.19 — The Parameter $\rho_{X,Y}$ Is the Correlation Coefficient

For bivariate Gaussian  $X, Y$  with PDF given by Def. 5.10, the parameter  $\rho_{X,Y}$  equals the correlation coefficient of  $X$  and  $Y$  as defined in Def. 5.6.

---



---



---



---

★ **Theorem 5.20 — Uncorrelated  $\Leftrightarrow$  Independent (Bivariate Gaussian Only!) ★ [PROFESSOR-FLAGGED — MEMORIZE]**

Bivariate Gaussian random variables  $X, Y$  are uncorrelated **if and only if** they are independent. (Setting  $\rho_{X,Y} = 0$  in Def. 5.10 factors the PDF into  $f_X(x)f_Y(y)$ .) This is the **one exception** to Theorem 5.17(c)'s one-way implication.

**Theorem 5.21 — Linear Combinations of Jointly Gaussian Variables**

If  $X, Y$  are bivariate Gaussian and  $W_1 = a_1X + b_1Y$ ,  $W_2 = a_2X + b_2Y$  (linearly independent combinations), then  $W_1, W_2$  are *also* bivariate Gaussian, with

$$\mathbb{E}[W_i] = a_i\mu_X + b_i\mu_Y, \quad \text{Var}[W_i] = a_i^2\sigma_X^2 + b_i^2\sigma_Y^2 + 2a_ib_i\rho_{X,Y}\sigma_X\sigma_Y,$$

$$\text{Cov}[W_1, W_2] = a_1a_2\sigma_X^2 + b_1b_2\sigma_Y^2 + (a_1b_2 + a_2b_1)\rho_{X,Y}\sigma_X\sigma_Y.$$

Even the partial fact “ $W_i$  alone is Gaussian” is powerful — once you know that, only means/variances/covariances remain to be computed.

**Example 5.19–5.20 — PDF of  $Y = X + Z$  and the joint PDF of  $(X, Y)$** 

$X \sim \text{Gaussian}(0, \sigma_X)$ ,  $Z \sim \text{Gaussian}(0, \sigma_Z)$  independent. By Thm 5.21,  $Y = X + Z$  is  $\text{Gaussian}(0, \sqrt{\sigma_X^2 + \sigma_Z^2})$ . With  $\sigma_X = 4, \sigma_Z = 3$ :  $\sigma_Y = 5$ ,  $\rho_{X,Y} = \sqrt{(\sigma_X^2/\sigma_Z^2)/(1 + \sigma_X^2/\sigma_Z^2)} = 4/5$ , and substituting into Def. 5.10:

$$f_{X,Y}(x, y) = \frac{1}{24\pi} e^{-(25x^2/16 - 2xy + y^2)/18}.$$

## Part III

## Chapter 6 — Probability Models of Derived Random Variables

## 7 Functions Yielding Continuous Random Variables (§6.2)

**The General Method — the single most important idea in Chapter 6**

To get the PDF of  $W = g(X)$ : **(1)** find the CDF  $F_W(w) = P[g(X) \leq w]$  by identifying which  $x$  satisfy  $g(x) \leq w$  and integrating  $f_X$  over that region; **(2)** differentiate:  $f_W(w) = dF_W(w)/dw$ . This always works, and is the *only* reliable method when  $g$  is not monotonic (e.g.  $Y = X^2$ ,  $Y = cX^2$ , limiters). It is *not* the fastest way to get  $E[g(X)]$  alone — for that, use Thm 4.4 directly on  $f_X$ .

**Theorem 6.2 — Scaling:  $W = aX$ ,  $a > 0$** 

$$F_W(w) = F_X(w/a), \quad f_W(w) = \frac{1}{a} f_X(w/a).$$

**Theorem 6.3 — Scaling Within Standard Families ( $W = aX$ ,  $a > 0$ )**

| If $X$ is . . .           | then $W = aX$ is . . .      |
|---------------------------|-----------------------------|
| uniform( $b, c$ )         | uniform( $ab, ac$ )         |
| exponential( $\lambda$ )  | exponential( $\lambda/a$ )  |
| Erlang( $n, \lambda$ )    | Erlang( $n, \lambda/a$ )    |
| Gaussian( $\mu, \sigma$ ) | Gaussian( $a\mu, a\sigma$ ) |

**Theorem 6.4 — Shift:  $W = X + b$** 

$$F_W(w) = F_X(w - b), \quad f_W(w) = f_X(w - b).$$

Combined with Thm 6.2: for  $W = aX + b$  ( $a > 0$ ),  $f_W(w) = \frac{1}{a} f_X\left(\frac{w - b}{a}\right)$ .

★ **Theorem 6.4 (Example) —  $W = X^2$  — the archetype for EVERY power/energy problem ( $Y = cX^2$ )** ★ [PROFESSOR-FLAGGED — MEMORIZE]

$X \sim \text{uniform}(-1, 3)$ :  $f_X(x) = 1/4$  on  $(-1, 3)$ . Since  $W = X^2 \geq 0$  always and  $X \in (-1, 3) \Rightarrow W \in [0, 9)$ :

$$F_W(w) = P[X^2 \leq w] = P[-\sqrt{w} \leq X \leq \sqrt{w}] = \int_{-\sqrt{w}}^{\sqrt{w}} f_X(x) dx,$$

**but clip the lower limit to  $X$ 's actual support:** for  $0 \leq w < 1$ ,  $-\sqrt{w} > -1$  so both limits are valid  $\Rightarrow F_W(w) = \sqrt{w}/2$ . For  $1 \leq w < 9$ ,  $-\sqrt{w} \leq -1$  falls *outside* the support, so the lower limit clips to  $-1$ :  $F_W(w) = (\sqrt{w} + 1)/4$ . Differentiating piecewise:

$$f_W(w) = \begin{cases} \frac{1}{4\sqrt{w}} & 0 < w < 1 \\ \frac{1}{8\sqrt{w}} & 1 < w < 9 \\ 0 & \text{otherwise.} \end{cases}$$

**This is exactly the method for “current  $X$  uniform, power  $Y = cX^2$ ” problems** (see Exam 2, Problem 3): (i) note  $Y \geq 0$ , get  $Y$ 's range from  $X$ 's range; (ii) write  $F_Y(y) = P[-\sqrt{y/c} \leq X \leq \sqrt{y/c}]$ ; (iii) integrate  $f_X$  over that interval, **clipping to  $X$ 's actual support**; (iv) differentiate piecewise.

### Theorem 6.5 — Inverse-CDF (Probability Integral Transform) Method

Let  $U \sim \text{uniform}(0, 1)$  and  $F(x)$  an invertible CDF. Then  $X = F^{-1}(U)$  has CDF  $F_X(x) = F(x)$ . (Useful for simulation; requires  $F$  to be easily invertible — fails for Gaussian and for mixed random variables.)

### Example 6.4–6.6 — Generating standard families from Uniform(0, 1)

**Exponential:** solve  $u = 1 - e^{-x} \Rightarrow x = -\ln(1 - u)$ , so  $X = -\ln(1 - U)$  is exponential(1).  
**Uniform( $a, b$ ):**  $X = a + (b - a)U$ .

**Quiz 6.2 (Rayleigh):** If  $X \sim \text{exponential}(\lambda)$ , then  $Y = \sqrt{X}$  is Rayleigh:  $F_Y(y) = P[X \leq y^2] =$

$1 - e^{-\lambda y^2} \Rightarrow f_Y(y) = 2\lambda y e^{-\lambda y^2}$ . Matching to the Rayleigh( $a$ ) form  $f(y) = (y/a^2)e^{-y^2/2a^2}$  gives  
 $a = 1/\sqrt{2\lambda}$ .

---

## 8 Functions Yielding Discrete or Mixed Random Variables (§6.3)

### Setup

A hard limiter (clipper) turns a continuous input into a **discrete** output; a soft limiter turns it into a **mixed** output (continuous body + impulses where the input was clipped). This is exactly the technique needed for range-limited measurement problems (see Exam 2, Problem 3(c)).

### Example 6.7 — Hard limiter: continuous $\rightarrow$ discrete

$g(x) = 1$  if  $x < 0$ ,  $g(x) = 3$  if  $x > 0$ . Then  $Y \in \{1, 3\}$  is discrete even though  $X$  is continuous:  $F_Y(1) = P[X < 0] = F_X(0)$ , so

$$F_Y(y) = \begin{cases} 0 & y < 1 \\ F_X(0) & 1 \leq y < 3, \\ 1 & y \geq 3 \end{cases}, \quad f_Y(y) = F_X(0) \delta(y - 1) + [1 - F_X(0)] \delta(y - 3).$$

### Example 6.8 — Soft limiter: continuous $\rightarrow$ mixed (the impulse-at-the-clip trick)

Microphone voltage  $V \sim \text{Gaussian}(0, 5)$ , clipped at  $\pm 10$ :  $W = g(V) = -10$  if  $V < -10$ ;  $W = V$  if  $-10 \leq V \leq 10$ ;  $W = 10$  if  $V > 10$ . For  $-10 \leq w < 10$ :  $F_W(w) = F_V(w) = \Phi(w/5)$ . The CDF *jumps* at  $w = \pm 10$  (weight  $1 - \Phi(2) \approx 0.0228$  at each end, by symmetry), so the PDF is **mixed**:

$$f_W(w) = \underbrace{0.023 \delta(w + 10)}_{\text{point mass at floor}} + \underbrace{\frac{1}{5\sqrt{2\pi}} e^{-w^2/50}}_{-10 < w < 10} + \underbrace{0.023 \delta(w - 10)}_{\text{point mass at ceiling}}.$$

**General recipe for “range-limited” problems:** wherever the limiter is inactive, the PDF is just  $f_X$ ; wherever it clips to a constant  $c$ , add an impulse of weight = the tail probability being clipped away, i.e.  $[1 - F_X(\text{clip point})] \delta(w - c)$ .

## Part IV

## Chapter 13 — Stochastic (Random) Processes

## 9 Setup: Mean and Autocorrelation of a Process (§13.7)

**Scope Note**

Brownian motion (§13.6) and the Poisson process (§13.4–13.5) are **not** exam topics — you do not need their definitions, properties, or derivations, and no counter-example built from either process is required. Focus entirely on the positive content of §13.7–13.11 below: the definitions and theorems that let you classify and work with a *given* process (mean, autocorrelation, stationarity, WSS, ergodicity, cross-correlation, Gaussian processes).

**Definition 13.11 — Mean Function of a Process**

$$\mu_X(t) = \mathbb{E}[X(t)].$$

**Definition 13.12 / 13.13 — Autocovariance and Autocorrelation**

$$C_X(t, \tau) = \text{Cov}[X(t), X(t + \tau)], \quad R_X(t, \tau) = \mathbb{E}[X(t)X(t + \tau)].$$

At  $\tau = 0$ :  $C_X(t, 0) = \text{Var}[X(t)]$  and  $R_X(t, 0) = \mathbb{E}[X^2(t)]$  (**average power**).

**Theorem 13.9 — Autocovariance from Autocorrelation**

$$C_X(t, \tau) = R_X(t, \tau) - \mu_X(t)\mu_X(t + \tau).$$

(This is just Theorem 5.16(a) applied at two different time instants.) The full worked illustration of  $\mu_X(t)$  and  $R_X(t, \tau)$  for a concrete process is Example 13.22 in the next section — the random-phase AM carrier — which is the professor's flagship example for this whole part of the chapter.

## 10 Stationary Processes (§13.8)

### Definition 13.14 — Stationary Process

$X(t)$  is stationary iff for all time instants  $t_1, \dots, t_m$  and any  $\tau$ :

$$f_{X(t_1), \dots, X(t_m)}(x_1, \dots, x_m) = f_{X(t_1+\tau), \dots, X(t_m+\tau)}(x_1, \dots, x_m).$$

Every joint PDF must be shift-invariant — not just the first-order PDF.

### ★ Theorem 13.10 — Linear Transformations Preserve Stationarity ★ [PROFESSOR-FLAGGED — MEMORIZE]

#### ★ Professor's review sheet, item 1 of 3.

Let  $X(t)$  be a stationary process. For constants  $a > 0$  and  $b$ ,  $Y(t) = aX(t) + b$  is also a stationary process.

*Proof.* For any  $t_1, \dots, t_n$ ,  $f_{Y(t_1), \dots, Y(t_n)}(y_1, \dots, y_n) = a^{-n} f_{X(t_1), \dots, X(t_n)}\left(\frac{y_1-b}{a}, \dots, \frac{y_n-b}{a}\right)$ . Since  $X(t)$  is stationary, the right side is unchanged by shifting every  $t_i \rightarrow t_i + \tau$ , so the left side is too — hence  $Y(t)$  is stationary.  $\square$

### ★ Theorem 13.11 — Three Consequences of Stationarity ★ [PROFESSOR-FLAGGED — MEMORIZE]

#### ★ Professor's review sheet, item 2 of 3.

For a stationary process  $X(t)$ , for *all*  $t$ :

$$(a) \mu_X(t) = \mu_X \quad (\text{constant mean}), \quad (b) R_X(t, \tau) = R_X(\tau) \quad (\text{depends on } \tau \text{ only}),$$

$$(c) C_X(t, \tau) = R_X(\tau) - \mu_X^2 = C_X(\tau).$$

*Proof.* Stationarity  $\Rightarrow f_{X(t)} = f_{X(0)}$  for all  $t$ , so  $\mu_X(t) = \int x f_{X(0)}(x) dx \equiv \mu_X$  for all  $t$ . Likewise  $f_{X(t), X(t+\tau)} = f_{X(0), X(\tau)}$  gives  $R_X(t, \tau) = R_X(0, \tau) \equiv R_X(\tau)$ . Part (c) follows from Theorem 13.9.  $\square$

**Key takeaway:** To show a process is *stationary*, verify all three parts of Theorem 13.11 directly from the process's definition: constant mean,  $\tau$ -only autocorrelation, and  $\tau$ -only autocovariance. Example 13.22 right below does exactly this for a concrete process.

★ **Theorem 13.22 — Random-Phase AM Carrier — THE professor's key worked example** ★ [PROFESSOR-FLAGGED — MEMORIZE]

★ **The exact example flagged on the review sheet.**

$X(t) = A \cos(2\pi f_c t + \Theta)$ , where  $\Theta \sim \text{Uniform}(0, 2\pi)$ :

$$f_{\Theta}(\theta) = \begin{cases} 1/(2\pi) & 0 \leq \theta < 2\pi \\ 0 & \text{otherwise} \end{cases}, \quad \int_0^{2\pi} f_{\Theta}(\theta) d\theta = 1.$$

**Key lemma** (used twice): for fixed  $a$  and nonzero integer  $k$ ,

$$\mathbb{E}[\cos(a + k\Theta)] = \int_0^{2\pi} \cos(a + k\theta) \frac{d\theta}{2\pi} = \frac{\sin(a + 2\pi k) - \sin a}{2\pi k} = 0 \quad (\text{since } \sin(a + 2\pi k) = \sin a).$$

**Mean** (take  $a = 2\pi f_c t$ ,  $k = 1$ ):  $\mu_X(t) = A \mathbb{E}[\cos(2\pi f_c t + \Theta)] = 0$ .

**Autocorrelation:** using  $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$  with  $A - B = -2\pi f_c \tau$  (constant) and  $A + B = 2\pi f_c(2t + \tau) + 2\Theta$ :

$$R_X(t, \tau) = \frac{A^2}{2} \mathbb{E}[\cos(2\pi f_c \tau) + \cos(2\pi f_c(2t + \tau) + 2\Theta)] = \frac{A^2}{2} \cos(2\pi f_c \tau)$$

(the second term vanishes by the lemma with  $k = 2$ ). So

$$R_X(t, \tau) = \frac{A^2}{2} \cos(2\pi f_c \tau) = R_X(\tau).$$

**Conclusion:** constant zero mean +  $\tau$ -only autocorrelation  $\Rightarrow X(t)$  is **wide-sense stationary** (formally classified in Example 13.23, using Definition 13.15 below).

## 11 Wide Sense Stationary Processes & Ergodicity (§13.9)

### Definition 13.15 — Wide Sense Stationary (WSS)

$X(t)$  is WSS iff for all  $t$ :

$$E[X(t)] = \mu_X \quad (\text{constant}), \quad R_X(t, \tau) = R_X(0, \tau) = R_X(\tau) \quad (\tau \text{ only}).$$

Every *stationary* process is WSS (Thm 13.11); the converse need not hold — so WSS is the strictly larger, easier-to-check class. This is the property actually used on exams.

### Example 13.23 — The AM carrier of Example 13.22 is WSS

$\mu_X(t) = 0$  (constant) and  $R_X(t, \tau) = (A^2/2) \cos(2\pi f_c \tau)$  (depends on  $\tau$  only)  $\Rightarrow X(t)$  is **WSS** by Definition 13.15.

### ★ Theorem 13.12 — Properties of the WSS Autocorrelation Function ★ [PROFESSOR-FLAGGED — MEMORIZE]

★ Professor's review sheet, item 3 of 3.

For a WSS process  $X(t)$ :

$$R_X(0) \geq 0, \quad R_X(\tau) = R_X(-\tau) \quad (\text{even}), \quad R_X(0) \geq |R_X(\tau)| \quad (\text{peak at } \tau=0).$$

*Proof.* (1)  $R_X(0) = E[X^2(t)] \geq 0$ . (2) Substituting  $u = t + \tau$ :  $R_X(t, \tau) = R_X(u, -\tau) = R_X(-\tau)$  by WSS. (3)  $\text{Var}[X(t)] = C_X(0)$  is constant (WSS); by the Cauchy–Schwarz-type bound of Theorem 5.14,  $C_X(\tau) \leq C_X(0)$ , and adding  $\mu_X^2$  to both sides of a squared inequality gives  $R_X(\tau)^2 \leq R_X(0)^2$ .  $\square$

**Key takeaway:** Use this theorem to test whether a *candidate function* could possibly be a valid autocorrelation: it must be even, and its value at  $\tau = 0$  must dominate every other value. Chapter 13 Homework Problem 13.9.1 (below) is built entirely around this check.

**Definition 13.16 — Average Power**

The average power of a WSS process is  $R_X(0) = \mathbb{E}[X^2(t)]$  (physically: power dissipated by a signal  $x(t)$  across a  $1\Omega$  resistor).

**11.1 Ergodicity and Time Averages**

★ **Theorem Concept — Ergodicity — “Section 13.9” as flagged by the professor** ★  
**[PROFESSOR-FLAGGED — MEMORIZE]**

★ **A wide sense stationary process is *ergodic* if statistical (ensemble) averages such as  $\mathbb{E}[X(t)]$  and  $\mathbb{E}[X^2(t)]$  equal the corresponding *time* averages of a single sample function.**

Time average of one sample function over duration  $2T$  (the measurable “DC value”):

$$\bar{X}(T) = \frac{1}{2T} \int_{-T}^T x(t) dt, \quad \overline{X^2}(T) = \frac{1}{2T} \int_{-T}^T x^2(t) dt \quad (\text{time-avg. power}).$$

$X(t)$  is **ergodic** if  $\lim_{T \rightarrow \infty} \bar{X}(T) = \mu_X$  (mean-ergodic) and similarly for other statistics:

$$\langle X(t) \rangle \equiv \lim_{T \rightarrow \infty} \bar{X}(T) = \mathbb{E}[X(t)] = \mu_X, \quad \langle X^2(t) \rangle \equiv \lim_{T \rightarrow \infty} \overline{X^2}(T) = \mathbb{E}[X^2(t)].$$

**Why it matters:** for an ergodic process,  $\mu_X$ ,  $\mathbb{E}[X^2(t)]$ , and other ensemble averages can be *measured* with an ordinary voltmeter on a single sample path — you never need to repeat the whole random experiment many times.

**Theorem 13.13 — Mean-Ergodic Unbiased/Consistent Estimator**

Let  $X(t)$  be stationary with mean  $\mu_X$  and autocovariance  $C_X(\tau)$ . If  $\int_{-\infty}^{\infty} |C_X(\tau)| d\tau < \infty$ , then  $\bar{X}(T)$  is an **unbiased** ( $\mathbb{E}[\bar{X}(T)] = \mu_X$ ) and **consistent** ( $\lim_{T \rightarrow \infty} \text{Var}[\bar{X}(T)] = 0$ ) estimator of  $\mu_X$ .

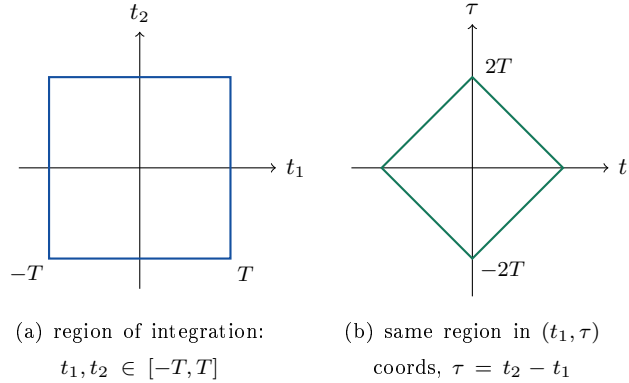
*Proof. Unbiased:*  $\mathbb{E}[\bar{X}(T)] = \frac{1}{2T} \int_{-T}^T \mathbb{E}[X(t)] dt = \mu_X$ . **Consistent:**  $\text{Var}[\bar{X}(T)] = \frac{1}{4T^2} \iint_{-T}^T C_X(t' - t) dt' dt \leq \frac{2T \cdot K}{4T^2} = \frac{K}{2T} \rightarrow 0$ , where  $K = \int_{-\infty}^{\infty} |C_X(\tau)| d\tau < \infty$  bounds the inner integral.  $\square$

### Alternative (Chebyshev) Route to the Same Result — Supplementary Notes

An equivalent presentation (from supplementary course notes) phrases mean-ergodicity via Chebyshev's inequality:  $P[|\bar{X}(T) - E[\bar{X}(T)]| \leq \varepsilon] \geq 1 - \text{Var}[\bar{X}(T)]/\varepsilon^2$ . Taking  $T \rightarrow \infty$ : if  $\text{Var}[\bar{X}(T)] \rightarrow 0$ , the right side  $\rightarrow 1$ , so  $\bar{X}(T) \rightarrow \mu_X$  with probability 1. A full change-of-variables ( $\tau = t_2 - t_1$ ) computation gives the sharper closed form

$$\text{Var}[\bar{X}(T)] = \frac{1}{T} \int_0^{2T} \left(1 - \frac{|\tau|}{2T}\right) C_X(\tau) d\tau,$$

with sufficient condition  $\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-2T}^{2T} |C_X(\tau)| d\tau = 0$  — guaranteed whenever  $\int_{-\infty}^{\infty} |C_X(\tau)| d\tau < \infty$ , matching Theorem 13.13.



### Example Ergodicity worked examples (supplementary) —

$X(t) = \cos(\omega t + \Theta)$ ,  $\Theta \sim \text{Uniform}(0, 2\pi)$ : mean = 0 (constant),  $R_X(\tau) = \frac{1}{2} \cos \omega \tau$  from both the ensemble computation and the time-average computation  $\langle X(t)X(t+\tau) \rangle = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \cos(\omega t + \theta) \cos(\omega t + \omega \tau + \theta) dt = \frac{1}{2} \cos \omega \tau \Rightarrow$  **ergodic**.

$R_X(\tau) = 25 + \frac{4}{1 + 6\tau^2}$  (stationary, ergodic, no periodic component):  $\mu_X^2 = \lim_{\tau \rightarrow \infty} R_X(\tau) = 25 \Rightarrow \mu_X = 5$ ;  $E[X^2(t)] = R_X(0) = 29 \Rightarrow \text{Var}[X(t)] = 29 - 25 = 4$ .

$R_X(\tau) = A\delta(\tau)$ , zero mean:  $\sigma_T^2 = A/T \rightarrow 0$  as  $T \rightarrow \infty \Rightarrow$  **mean-ergodic**.

**Random binary transmission**,  $R(\tau) = 1 - |\tau|/T$ :  $\text{Var}[\bar{X}_T] = 2/3 \neq 0$  even as  $T \rightarrow \infty \Rightarrow$  **NOT mean-ergodic** (a triangular autocorrelation that never decays relative to the averaging window fails the ergodic condition).

**Key takeaway:** Exam pattern: you'll be given  $R_X(\tau)$  and asked either (i) to find  $\mu_X, \text{Var}[X(t)]$  using  $\mu_X^2 = \lim_{|\tau| \rightarrow \infty} R_X(\tau)$  and  $R_X(0) = \text{E}[X^2(t)]$ , or (ii) to decide ergodicity by checking whether  $\text{Var}[\bar{X}(T)] \rightarrow 0$ .

---

### Example Valid autocorrelation functions? (quiz-style) — A

ny candidate  $R(\tau)$  **must** be even ( $R(\tau) = R(-\tau)$ ) and satisfy  $|R(\tau)| \leq R(0)$  (Theorem 13.12). A function like  $R(\tau) = e^{-\tau^2} \sin \tau$  is **odd** (sin is odd,  $e^{-\tau^2}$  is even, product is odd)  $\Rightarrow$  **not** a valid autocorrelation function — purely even building blocks ( $e^{-|\tau|}$ ,  $e^{-\tau^2}$ ,  $\cos \tau$ , and products thereof) survive this test.

---

## 12 Cross-Correlation (§13.10)

---

### Definition 13.17 — Cross-Correlation Function

$$R_{XY}(t, \tau) = \mathbb{E}[X(t)Y(t + \tau)].$$


---

---

### Definition 13.18 — Jointly Wide Sense Stationary

$X(t), Y(t)$  are jointly WSS if both are individually WSS *and*  $R_{XY}(t, \tau) = R_{XY}(\tau)$  depends only on  $\tau$ .

---



---

### Theorem 13.14 — Symmetry of Cross-Correlation

If  $X(t), Y(t)$  are jointly WSS:  $R_{XY}(\tau) = R_{YX}(-\tau)$ .

---



---

### Example 13.24 — Noisy observation $Y(t) = X(t) + N(t)$ : three WSS/jointly-WSS checks at once

$X(t), N(t)$  independent WSS,  $\mu_N = 0$ . Then  $\mathbb{E}[Y(t)] = \mu_X$  (constant) and

$$R_Y(t, \tau) = \mathbb{E}[[X(t) + N(t)][X(t + \tau) + N(t + \tau)]] = R_X(\tau) + R_N(\tau)$$

(cross terms vanish by independence and  $\mu_N = 0$ )  $\Rightarrow Y(t)$  is **WSS**. Also  $R_{YX}(t, \tau) = R_X(\tau) \Rightarrow X, Y$  **jointly WSS**; and  $R_{YN}(t, \tau) = R_N(\tau) \Rightarrow Y, N$  **jointly WSS**. (*This is the standard “signal plus independent noise” template — expect a variant of it on the exam.*)

---

**Key takeaway:** Being individually WSS and being *jointly* WSS with another process are two separate checks — always verify both from the definitions rather than assuming one implies the other.

## 13 Gaussian Processes (§13.11)

### Definition 13.19 — Gaussian Process

$X(t)$  is a Gaussian process iff  $\mathbf{X} = [X(t_1) \cdots X(t_k)]'$  is a Gaussian random vector for *every*  $k$  and every choice of  $t_1, \dots, t_k$ .

### Theorem 13.15 — A WSS Gaussian Process Is Stationary

If  $X(t)$  is a WSS Gaussian process, then  $X(t)$  is (strict-sense) stationary.

*Proof.* For any  $t_1, \dots, t_k$  and shift  $\tau$ , WSS gives equal means and (since  $C_X$  depends only on the lag  $t_j - t_i$ ) equal covariance matrices for  $\mathbf{X} = [X(t_i)]$  and  $\tilde{\mathbf{X}} = [X(t_i + \tau)]$ . A Gaussian vector's PDF is fully determined by its mean vector and covariance matrix, so  $f_{\mathbf{X}} = f_{\tilde{\mathbf{X}}}$  for arbitrary  $k$  — i.e. stationary.  $\square$

### Definition 13.20 — White Gaussian Noise

$W(t)$  is white Gaussian noise iff it is a stationary Gaussian process with  $\mu_W = 0$  and  $R_W(\tau) = \eta_0 \delta(\tau)$ . Consequence: samples at distinct times are *independent* Gaussians (uncorrelated + jointly Gaussian  $\Rightarrow$  independent, via Theorem 5.20). Average power  $R_W(0) = \eta_0 \delta(0) = \infty$  — physically impossible, but a useful idealization.

★ **Theorem Bivariate special case — The Gaussian-process joint PDF reduces to Definition 5.10** ★ [PROFESSOR-FLAGGED — MEMORIZE]

For a stationary Gaussian process, the joint PDF of any two samples  $X_1 = X(t)$ ,  $X_2 = X(t + \tau)$  is exactly the bivariate Gaussian PDF of §6 with  $\sigma_1 = \sigma_2 = \sqrt{R_X(0)}$  and  $\rho = C_X(\tau)/R_X(0)$ . **Worked example:**  $\mu_X(t) = 0$ ,  $R_X(\tau) = 2^{-|\tau|}$ . Then  $\text{Var}[X(t)] = R_X(0) = 1$ ,  $\text{Cov}[X(t), X(t+1)] = R_X(1) = 1/2 \Rightarrow \rho = 1/2$ , giving

$$f_{X(t), X(t+1)}(x_1, x_2) = \frac{1}{\pi\sqrt{3}} \exp\left(-\frac{2(x_1^2 - x_1x_2 + x_2^2)}{3}\right).$$

### 13.1 Master Comparison Table: Which Processes Are Stationary / WSS?

| Process                                                                          | Stationary?        | WSS?       | Why / why not                                                            |
|----------------------------------------------------------------------------------|--------------------|------------|--------------------------------------------------------------------------|
| Random-phase carrier<br>$A \cos(2\pi f_c t + \Theta)$ , $\Theta \sim U(0, 2\pi)$ | Not shown directly | <b>Yes</b> | $\mu_X = 0$ , $R_X(\tau) = \frac{A^2}{2} \cos 2\pi f_c \tau$ (Ex. 13.22) |
| White Gaussian noise                                                             | <b>Yes</b>         | <b>Yes</b> | Gaussian + WSS $\Rightarrow$ stationary (Thm 13.15)                      |
| $W(t) = X(t)Y(t)$ ,<br>$X, Y$ indep. WSS                                         | Not shown directly | <b>Yes</b> | $R_W(\tau) = R_X(\tau)R_Y(\tau)$ (HW 13.10.1)                            |
| $Y(t) = X(t - t_0)$ , $X$ WSS                                                    | Not shown directly | <b>Yes</b> | $R_Y(\tau) = R_X(\tau)$ : a delay doesn't change the ACF (HW 13.10.3)    |

## Part V

# Full Exam & Homework Problem Bank — Every Problem, Fully Worked

## How to Use This Part

The professor said the final reuses the exact material from Exam 1 and Exam 2. Every problem below is transcribed and solved completely, tagged with the theorem/section it exercises so you can jump back to the relevant box above. The Chapter 13 homework set (§13.9–13.11) is included as well, since it directly exercises the professor-flagged WSS/cross-correlation/ Gaussian-process theorems.

## 14 Exam 1 — Full Solutions (7 Problems)

**Exam 1, Problem 1 — Mixed random variable — normalize, sketch CDF, compute probabilities** [§3, Delta Functions/Mixed RVs]

Given

$$f_X(x) = \frac{1}{8}\delta(x+2) + \frac{1}{16}\delta(x+1) + \frac{1}{16}\delta(x-1) + \begin{cases} Ax^2 & |x| \leq 2 \\ 0 & \text{otherwise} \end{cases}.$$

(a) **Find A.** Impulse probability totals  $\frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4}$ , so the continuous part must supply  $\frac{3}{4}$ :

$$A \int_{-2}^2 x^2 dx = A \cdot \frac{16}{3} = \frac{3}{4} \implies \boxed{A = \frac{9}{64}}.$$

(b)/(c)/(d): **full CDF** (built case-by-case, adding each impulse's jump as  $x$  crosses it):

$$F_X(x) = \begin{cases} 0 & x < -2 \\ \frac{1}{2} + \frac{3x^3}{64} & -2 \leq x < -1 \\ \frac{9}{16} + \frac{3x^3}{64} & -1 \leq x < 1 \\ \frac{5}{8} + \frac{3x^3}{64} & 1 \leq x < 2 \\ 1 & x \geq 2. \end{cases}$$

So  $F_X(1) = \frac{5}{8} + \frac{3}{64} = \boxed{\frac{43}{64}}$ , and  $P[-1 < X \leq 2] = F_X(2) - F_X(-1) = 1 - \frac{33}{64} = \boxed{\frac{31}{64}}$ .

**Exam 1, Problem 2 — Joint PDF, marginals,  $E[X]$ ,  $E[X^2]$ ,  $\text{Var}[X]$** **[§4–5]**

$$f_{X,Y}(x,y) = \frac{x+y}{3} \text{ on } 0 \leq x \leq 1, 0 \leq y \leq 2.$$

$$f_X(x) = \int_0^2 \frac{x+y}{3} dy = \frac{2x+2}{3}, \quad 0 \leq x \leq 1.$$

$$E[X] = \int_0^1 x \cdot \frac{2x+2}{3} dx = \frac{5}{9}, \quad E[X^2] = \int_0^1 x^2 \cdot \frac{2x+2}{3} dx = \frac{7}{18},$$

$$\text{Var}[X] = \frac{7}{18} - \left(\frac{5}{9}\right)^2 = \boxed{\frac{13}{162}}.$$

**Exam 1, Problem 3 — Joint PMF table — marginals,  $E[Q]$ ,  $\text{Var}[G]$  [Joint/marginal PMF]**

Reconstructed joint PMF  $P_{Q,G}(q,g)$ : row sums  $P_Q(0) = 0.6, P_Q(1) = 0.4$ ; column sums  $P_G(0)=0.1, P_G(1)=0.3, P_G(2)=0.4, P_G(3)=0.2$ .

$$P[Q=0] = 0.6, \quad P[Q=G] = 0.18, \quad P[G=0] = 0.1,$$

$$P[G > 1] = 0.6, \quad P[G > Q] = 0.78.$$

$$E[Q] = 0.4. \quad E[G] = 1.7, \quad E[G^2] = 3.7 \implies \boxed{\text{Var}[G] = 3.7 - 1.7^2 = 0.81}.$$

**Exam 1, Problem 4 — Normalize a continuous PDF, CDF, probabilities CDF/PDF basics****[§1,**

$$f_X(x) = c(1-x^2) \text{ on } -1 \leq x \leq 1. \text{ Normalizing: } c \int_{-1}^1 (1-x^2) dx = \frac{4}{3}c = 1 \implies \boxed{c = \frac{3}{4}}.$$

$$F_X(x) = \frac{3}{4} \left[ (x+1) - \frac{1}{3}(x^3+1) \right].$$

$$P[X=0] = 0 \text{ (continuous r.v.)}; \quad F_X(0) = \frac{1}{2}; \quad P[0 < X < 0.5] = \frac{11}{32}; \quad P[|X - \frac{1}{2}| < \frac{1}{4}] = P\left[\frac{1}{4} < X < \frac{3}{4}\right] \approx 0.2734.$$

**Exam 1, Problem 5 — Bit-error probability: Gaussian noise,  $Q$ -function, SNR [§2]**

Bit-error probability =  $Q(\sqrt{E/\sigma^2})$ ,  $\text{SNR} = E/\sigma^2$ . (a)  $\text{SNR} = 4$ :  $Q(2) = 1 - \Phi(2) = 1 - 0.97725 = \boxed{0.02275}$ . (b) Need  $Q(\sqrt{\text{SNR}}) < 10^{-6} \Rightarrow \sqrt{\text{SNR}} \geq 4.75 \Rightarrow \boxed{\text{SNR} \geq (4.75)^2 = 22.56}$ .

**Exam 1, Problem 6 (Extra Credit) — Joint CDF — solve for a normalizing constant [CDF properties]**

$$F_{X,Y}(x, y) = a \left[ \frac{\pi}{2} + \tan^{-1}\left(\frac{x}{2}\right) \right] \left[ \frac{\pi}{2} + \tan^{-1}\left(\frac{y}{3}\right) \right].$$

Using  $\lim_{x,y \rightarrow \infty} F_{X,Y}(x, y) = 1$  and  $\tan^{-1}(\infty) = \pi/2$ :  $a\pi \cdot \pi = 1 \Rightarrow \boxed{a = \frac{1}{\pi^2}}$ . (Check:  $F(-\infty, y) = F(x, -\infty) = 0$  since  $\tan^{-1}(-\infty) = -\pi/2$ .)

**Exam 1, Problem 7 (Extra Credit) — Soft limiter — Jacobian / inverse-function method [§8]**

$Y = V(1 - e^{-X/a})$  for  $X \geq 0$ ;  $Y = -V(1 - e^{X/a})$  for  $X \leq 0$ . Solving each branch for  $x(y)$  and applying  $f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$ :

$$f_Y(y) = \frac{a}{V-y} f_X \left[ a \ln \left( \frac{V}{V-y} \right) \right] u(y) + \frac{a}{V+y} f_X \left[ -a \ln \left( \frac{V}{V+y} \right) \right] u(-y).$$

## 15 Exam 2 — Full Solutions (5 Problems)

**Exam 2, Problem 1 — Joint PDF moments and covariance** [§4–5] — textbook 5.7.12

$f_{X,Y}(x,y) = 4xy$  on the unit square.

$$E[X] = E[Y] = \frac{2}{3}, \quad E[X^2] = E[Y^2] = \frac{1}{2} \Rightarrow \text{Var}[X] = \text{Var}[Y] = \frac{1}{18}.$$

$$E[XY] = \int_0^1 \int_0^1 4x^2y^2 dy dx = \frac{4}{9} \Rightarrow \text{Cov}[X,Y] = \frac{4}{9} - \left(\frac{2}{3}\right)^2 = 0.$$

$$E[X+Y] = \frac{4}{3}, \quad \text{Var}[X+Y] = \frac{1}{18} + \frac{1}{18} + 2(0) = \boxed{\frac{1}{9}}.$$

**Exam 2, Problem 2 — Bivariate-Gaussian-type joint PDF** — back out  $\rho$ ,  $\sigma$ 's,  $c$  [§6] — textbook 5.9.5

$f_{X,Y}(x,y) = ce^{-(2x^2-4xy+4y^2)}$ . Matching the exponent to Definition 5.10's quadratic form:  $E[X] = E[Y] = 0$ ; solving the coefficient-matching equations gives  $\sigma_X = 1/\sqrt{2}$ ,  $\sigma_Y = 1/2$ ,  $\boxed{\rho = 1/\sqrt{2}}$ , and

$$c = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} = \boxed{\frac{2}{\pi}}.$$

Since  $\rho \neq 0$ :  $X, Y$  **are dependent** (by Theorem 5.20, since they're bivariate Gaussian, uncorrelated would have been equivalent to independent — but here they're correlated).

**Exam 2, Problem 3 — Power dissipation**  $Y = 9X^2$ , then range-limited  $W$  [§7–8] — textbook 6.3.10

$X \sim \text{Uniform}(-2, 2)$  (current),  $Y = 9X^2$  (power, Watts). Following the Example 6.4 template exactly (with  $c = 9$ ):  $0 \leq Y < 36$ , and for  $0 \leq y < 36$ ,

$$F_Y(y) = P\left[-\frac{\sqrt{y}}{3} \leq X \leq \frac{\sqrt{y}}{3}\right] = \frac{\sqrt{y}}{6} \implies f_Y(y) = \frac{1}{12\sqrt{y}}, \quad 0 < y \leq 36.$$

**Range-limited**  $W = Y$  if  $Y < 16$ , else  $W = 16$ : for  $0 \leq w < 16$ ,  $F_W(w) = F_Y(w) = \sqrt{w}/6$ ; at  $w = 16$  there's a jump from  $F_W(16^-) = 4/6$  to 1, i.e. an impulse of weight  $1 - \frac{4}{6} = \frac{1}{3}$ :

$$f_W(w) = \frac{1}{12\sqrt{w}} \quad (0 \leq w < 16) + \frac{1}{3}\delta(w - 16).$$

(This is the exact soft-limiter/impulse-weight technique of Example 6.8, applied on top of the  $Y=cX^2$  technique of Example 6.4.)

---

**Exam 2, Problem 4 — Joint PDF on a triangle — correlation and  $E[e^{X+Y}]$**     **[§5]** — textbook 5.8.8

$f_{X,Y}(x,y) = \frac{1}{2}$  on  $-1 \leq x \leq y \leq 1$ .

$$r_{X,Y} = E[XY] = \frac{1}{4} \int_{-1}^1 x(1-x^2) dx = \boxed{0} \quad (\text{odd integrand}).$$

$$E[e^{X+Y}] = \frac{1}{2} \int_{-1}^1 e^x(e - e^x) dx = \boxed{\frac{e^2}{4} + \frac{e^{-2}}{4} - \frac{1}{2}}.$$


---

**Exam 2, Problem 5 — Discrete joint PMF — full moment/covariance workup** **[§5]** — textbook 5.8.6

$P_{X,Y}(x,y) = 1/(4x)$  for integers  $1 \leq y \leq x \leq 4$ .

$$E[X] = \frac{5}{2}, \quad E[Y] = \frac{7}{4}, \quad \text{Var}[X] = \frac{5}{4}, \quad \text{Var}[Y] = \frac{41}{48}.$$

$$r_{X,Y} = E[XY] = 5 \implies \text{Cov}[X,Y] = 5 - \left(\frac{7}{4}\right)\left(\frac{10}{4}\right) = \frac{10}{16}.$$

$$\rho_{X,Y} = \frac{10/16}{\sqrt{(5/4)(41/48)}} \approx \boxed{0.605}.$$


---

## 16 Chapter 13 Homework — Full Solutions (6 Problems)

**Problem 13.9.1** — Which of the following are valid autocorrelation functions? [§11, Theorem 13.12]

$$R_1(\tau) = \delta(\tau), \quad R_2(\tau) = \delta(\tau) + 10, \quad R_3(\tau) = \delta(\tau - 10), \quad R_4(\tau) = \delta(\tau) - 10.$$

Test every candidate against Theorem 13.12: must be even ( $R(\tau) = R(-\tau)$ ), and (from the supplementary ergodic-process properties) if  $X(t)$  has nonzero mean and no periodic component,  $\lim_{|\tau| \rightarrow \infty} R_X(\tau) = \mu_X^2 \geq 0$ .

$R_1(\tau) = \delta(\tau)$ : even ( $\delta(\tau) = \delta(-\tau)$ ) — this is exactly the white-noise autocorrelation of Definition 13.20 (with  $\eta_0 = 1$ ). **Valid.**

$R_2(\tau) = \delta(\tau) + 10$ : even. It is the sum of two individually valid autocorrelations —  $\delta(\tau)$  (white noise) and the constant 10 (the autocorrelation of a “process” that is a random constant  $A$  with  $E[A^2] = 10$ , so  $R_A(\tau) = E[A \cdot A] = 10$  for all  $\tau$ ). The autocorrelation of a sum of two *independent* processes is the sum of their autocorrelations, so  $R_2$  is realizable. **Valid.**

$R_3(\tau) = \delta(\tau - 10)$ :  $R_3(-\tau) = \delta(-\tau - 10) = \delta(\tau + 10) \neq \delta(\tau - 10) = R_3(\tau)$  — **fails the evenness property outright. Not valid.**

$R_4(\tau) = \delta(\tau) - 10$ : even ( $\delta(\tau) - 10 = \delta(-\tau) - 10$ ), so it passes that test. But away from the origin ( $\tau \neq 0$ )  $R_4(\tau) = -10$  for *every* nonzero lag, so  $\lim_{|\tau| \rightarrow \infty} R_4(\tau) = -10$ . Since this limit must equal  $\mu_X^2 \geq 0$  for any real process,  $-10 < 0$  is impossible. **Not valid.**

Valid:  $R_1, R_2$ .      Not valid:  $R_3, R_4$ .

**Problem 13.9.5** — iid sequence  $X_n$ : find  $R_X[n, k]$  [Definition 13.13, Theorem 4.5(c)]

$X_n$  is iid with  $E[X_n] = \mu$ ,  $\text{Var}[X_n] = \sigma^2$ . Find  $R_X[n, k] = E[X_n X_{n+k}]$ .

**Case  $k = 0$ :**  $R_X[n, 0] = E[X_n^2] = \text{Var}[X_n] + (E[X_n])^2 = \sigma^2 + \mu^2$  (Theorem 4.5(c)).

**Case  $k \neq 0$ :**  $X_n$  and  $X_{n+k}$  are *independent* (different terms of an iid sequence), so by Theorem 5.17(b),  $R_X[n, k] = E[X_n] E[X_{n+k}] = \mu^2$ .

$$R_X[n, k] = \begin{cases} \sigma^2 + \mu^2 & k = 0 \\ \mu^2 & k \neq 0. \end{cases}$$

Independent of  $n$ : this confirms an iid sequence is (trivially) stationary, since every finite joint PDF is a product of identical marginals, hence shift-invariant.

**Problem 13.9.9 — WSS process modulated by an independent random phase [Example 13.22's lemma, Theorem 13.16]**

$X(t)$  is WSS with average power  $R_X(0) = E[X^2(t)] = 1$ .  $\Theta \sim \text{Uniform}(0, 2\pi)$ , independent of  $X(t)$ .

(a)  $E[X^2(t)] = R_X(0) = \boxed{1}$  (given directly — this is Definition 13.16's average power).

(b) Using Example 13.22's lemma with  $a = 2\pi f_c t$ ,  $k = 1$ :

$$E[\cos(2\pi f_c t + \Theta)] = \int_0^{2\pi} \cos(2\pi f_c t + \theta) \frac{d\theta}{2\pi} = \boxed{0}.$$

(c)  $Y(t) = X(t) \cos(2\pi f_c t + \Theta)$ . Since  $X(t)$  and  $\Theta$  are independent, Theorem 5.17(a) gives

$$E[Y(t)] = E[X(t)] \cdot E[\cos(2\pi f_c t + \Theta)] = E[X(t)] \cdot 0 = \boxed{0}$$

regardless of  $E[X(t)]$ .

(d) Average power of  $Y(t)$ :  $E[Y^2(t)] = E[X^2(t) \cos^2(2\pi f_c t + \Theta)]$ . Independence again gives  $E[Y^2(t)] = E[X^2(t)] \cdot E[\cos^2(2\pi f_c t + \Theta)]$ . Using  $\cos^2 \phi = \frac{1}{2}[1 + \cos 2\phi]$  and the lemma with  $a = 4\pi f_c t$ ,  $k = 2$ :

$$E[\cos^2(2\pi f_c t + \Theta)] = \frac{1}{2} + \frac{1}{2}E[\cos(4\pi f_c t + 2\Theta)] = \frac{1}{2} + 0 = \frac{1}{2}.$$

So average power  $= E[X^2(t)] \cdot \frac{1}{2} = 1 \cdot \frac{1}{2} = \boxed{1/2}$ .

**Problem 13.10.1 — Product of two independent WSS processes [Definitions 13.11–13.13, 13.17–13.18]****[Definitions**

$X(t), Y(t)$  independent WSS with means  $\mu_X, \mu_Y$  and autocorrelations  $R_X(\tau), R_Y(\tau)$ .  $W(t) = X(t)Y(t)$ .

(a) By independence (Theorem 5.17(a)):

$$\mu_W = E[W(t)] = E[X(t)]E[Y(t)] = \mu_X \mu_Y \quad (\text{constant}).$$

$$\begin{aligned} R_W(t, \tau) &= E[W(t)W(t+\tau)] = E[X(t)Y(t)X(t+\tau)Y(t+\tau)] \\ &= E[X(t)X(t+\tau)] \cdot E[Y(t)Y(t+\tau)] = R_X(\tau)R_Y(\tau), \end{aligned}$$

using independence of the  $X$  and  $Y$  processes to split the expectation of the product.  $R_W(t, \tau) = R_X(\tau)R_Y(\tau)$  depends only on  $\tau$ , and  $\mu_W$  is constant, so  $\boxed{W(t) \text{ is WSS.}}$

(b) Cross-correlation of  $W$  and  $X$ :

$$R_{WX}(t, \tau) = E[W(t)X(t+\tau)] = E[X(t)Y(t)X(t+\tau)] = E[X(t)X(t+\tau)] \cdot E[Y(t)] = R_X(\tau)\mu_Y,$$

using independence of  $Y(t)$  from  $X(t), X(t+\tau)$ . This depends only on  $\tau$ , and both  $W(t)$  and  $X(t)$  are individually WSS, so  $\boxed{\text{yes, jointly WSS.}}$

**Problem 13.10.3 — Autocorrelation/cross-correlation of a time-delayed WSS process**  
**[Definitions 13.13, 13.17–13.18]**

$X(t)$  WSS with  $R_X(\tau) = 10 \sin(2\pi \cdot 1000\tau)/(2\pi \cdot 1000\tau)$ .  $Y(t) = X(t - t_0)$ ,  $t_0 = 5 \times 10^{-5}\text{s}$ .

(a)  $R_Y(t, \tau) = E[Y(t)Y(t + \tau)] = E[X(t - t_0)X(t + \tau - t_0)]$ . The two arguments of  $X$  differ by exactly  $\tau$  (independent of  $t_0$ ), and  $X$  is WSS, so

$$R_Y(\tau) = R_X(\tau) = \frac{10 \sin(2\pi \cdot 1000\tau)}{2\pi \cdot 1000\tau}.$$

*Delaying a WSS process does not change its autocorrelation function.*

(b)  $R_{XY}(t, \tau) = E[X(t)Y(t + \tau)] = E[X(t)X(t + \tau - t_0)]$ . The two arguments of  $X$  now differ by  $\tau - t_0$ , so

$$R_{XY}(\tau) = R_X(\tau - t_0) = \frac{10 \sin(2\pi \cdot 1000(\tau - t_0))}{2\pi \cdot 1000(\tau - t_0)}.$$

(c) From (a),  $R_Y(\tau)$  depends only on  $\tau$ , and  $\mu_Y = E[X(t - t_0)] = \mu_X$  (constant, since  $X$  is WSS).  
 Yes,  $Y(t)$  is WSS.

(d) From (b),  $R_{XY}(t, \tau) = R_X(\tau - t_0)$  depends only on  $\tau$  (not on  $t$ ), and both  $X(t), Y(t)$  are individually WSS. Yes, jointly WSS.

**Problem 13.11.2 — Which transformations of a Gaussian process stay Gaussian?** [Definition 13.19, Theorem 5.21/8.11]

Given a Gaussian process  $X(t)$ , identify which are Gaussian processes: (a)  $2X(t)$ , (b)  $X(t/2)$ , (c)  $X(t)/2$ , (d)  $X(t) - X(t - 1)$ , (e)  $X(2t)$ .

By Definition 13.19, a process  $Z(t)$  is Gaussian iff  $[Z(t_1), \dots, Z(t_k)]'$  is a Gaussian random vector for *every* choice of  $k$  time instants. Two facts do all the work: (i) **re-indexing time** (evaluating  $X$  at a different, still-arbitrary set of time instants) trivially preserves the “any finite sample vector is Gaussian” property; (ii) a **linear transformation** of a Gaussian vector is again a Gaussian vector (Theorem 5.21/8.11).

(a)  $2X(t)$ : for any  $t_1, \dots, t_k$ ,  $[2X(t_1), \dots, 2X(t_k)]'$  is a linear (scalar) transform of the Gaussian vector  $[X(t_1), \dots, X(t_k)]'$ . Gaussian.

(b)  $X(t/2)$ : for any  $t_1, \dots, t_k$ ,  $[X(t_1/2), \dots, X(t_k/2)]'$  is simply a sample of  $X$  at the (still arbitrary) time instants  $t_1/2, \dots, t_k/2$  — Gaussian by Definition 13.19 directly. Gaussian.

(c)  $X(t)/2$ : same as (a) with scalar  $1/2$ . Gaussian.

(d)  $X(t) - X(t - 1)$ : for any  $t_1, \dots, t_k$ ,  $[X(t_1) - X(t_1 - 1), \dots, X(t_k) - X(t_k - 1)]'$  is a linear transformation of the  $2k$ -dimensional Gaussian vector  $[X(t_1), X(t_1 - 1), \dots, X(t_k), X(t_k - 1)]'$  (itself Gaussian, being  $2k$  samples of  $X$ ). Gaussian.

(e)  $X(2t)$ : same reasoning as (b), sampling  $X$  at times  $2t_1, \dots, 2t_k$ . Gaussian.

**All five are Gaussian processes.** *Lesson: the Gaussian-process property is extremely robust — it survives any linear operation on the values and any re-indexing (including nonlinear re-indexing like  $t \mapsto 2t$ ) of the time axis.*

---

## Part VI

## Formula Cheat Sheet

## Master Formula Reference (all chapters, one place)

| Topic                                        | Formula                                                                                                                                                                                                                                                                          |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDF                                          | $F_X(x) = P[X \leq x]; P[x_1 < X \leq x_2] = F_X(x_2) - F_X(x_1)$                                                                                                                                                                                                                |
| PDF                                          | $f_X(x) = dF_X/dx; \int f_X = 1; F_X(x) = \int_{-\infty}^x f_X$                                                                                                                                                                                                                  |
| Expectation                                  | $E[X] = \int x f_X(x) dx; E[g(X)] = \int g(x) f_X(x) dx$                                                                                                                                                                                                                         |
| Variance                                     | $\text{Var}[X] = E[X^2] - (E[X])^2; \text{Var}[aX + b] = a^2 \text{Var}[X]$                                                                                                                                                                                                      |
| Gaussian PDF                                 | $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}; F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$                                                                                                                                                                  |
| Q-function                                   | $Q(z) = 1 - \Phi(z) = P[Z > z]; \Phi(-z) = 1 - \Phi(z)$                                                                                                                                                                                                                          |
| Delta/sifting                                | $\int g(x)\delta(x-x_0)dx = g(x_0); \text{discrete PDF} = \sum P_X(x_i)\delta(x-x_i)$                                                                                                                                                                                            |
| Mixed RV                                     | PDF has impulses <i>and</i> nonzero finite values                                                                                                                                                                                                                                |
| $E[g(X, Y)]$                                 | $\iint g(x, y) f_{X,Y}(x, y) dx dy$                                                                                                                                                                                                                                              |
| $E[X + Y]$                                   | $= E[X] + E[Y]$ always (even if dependent)                                                                                                                                                                                                                                       |
| $\text{Var}[X + Y]$                          | $= \text{Var}[X] + \text{Var}[Y] + 2\text{Cov}[X, Y]$ — needs the joint model                                                                                                                                                                                                    |
| Covariance                                   | $\text{Cov}[X, Y] = E[XY] - E[X]E[Y]$                                                                                                                                                                                                                                            |
| Correlation                                  | $r_{X,Y} = E[XY]$                                                                                                                                                                                                                                                                |
| Correlation coeff.                           | $\rho_{X,Y} = \text{Cov}[X, Y] / (\sigma_X \sigma_Y) \in [-1, 1]$                                                                                                                                                                                                                |
| Independent $\Rightarrow$                    | $\text{Cov}[X, Y] = \rho_{X,Y} = 0$ (uncorrelated) — converse false in general                                                                                                                                                                                                   |
| Bivariate Gaussian                           | $f_{X,Y}(x, y) = \frac{\exp[-Q(x, y)/2(1-\rho^2)]}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}}, \quad \text{where } Q(x, y) = \left(\frac{x-\mu_X}{\sigma_X}\right)^2 - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2$ (full form: see §6) |
| Bivar. Gaussian, unc. $\Leftrightarrow$ ind. | only family where this equivalence holds                                                                                                                                                                                                                                         |
| $F_W(w), W = g(X)$                           | $F_W(w) = P[g(X) \leq w]$ , then differentiate                                                                                                                                                                                                                                   |
| $W = aX + b$                                 | $f_W(w) = \frac{1}{ a } f_X\left(\frac{w-b}{a}\right)$                                                                                                                                                                                                                           |
| $W = X^2$ (or $cX^2$ )                       | $F_W(w) = P[-\sqrt{w/c} \leq X \leq \sqrt{w/c}]$ , clip to $X$ 's support, differentiate piecewise                                                                                                                                                                               |
| Range-limited/clipped $W$                    | continuous part unchanged where active; impulse of weight = clipped tail probability at each clip point                                                                                                                                                                          |
| Mean function                                | $\mu_X(t) = E[X(t)]$                                                                                                                                                                                                                                                             |
| Autocorrelation                              | $R_X(t, \tau) = E[X(t)X(t+\tau)]; R_X(t, 0) = E[X^2(t)]$ (power)                                                                                                                                                                                                                 |
| Autocovariance                               | $C_X(t, \tau) = R_X(t, \tau) - \mu_X(t)\mu_X(t+\tau)$                                                                                                                                                                                                                            |
| Stationary                                   | every joint PDF is shift-invariant in $t$                                                                                                                                                                                                                                        |
| WSS                                          | $\mu_X(t) = \mu_X$ (const.) and $R_X(t, \tau) = R_X(\tau)$                                                                                                                                                                                                                       |
| WSS autocorrelation                          | $R_X(0) \geq 0; R_X(\tau) = R_X(-\tau); R_X(0) \geq  R_X(\tau) $                                                                                                                                                                                                                 |
| Ergodic (mean)                               | $\lim_{T \rightarrow \infty} \bar{X}(T) = \mu_X = E[X(t)]$ and similarly for $E[X^2(t)]$                                                                                                                                                                                         |
| Cross-correlation                            | $R_{XY}(t, \tau) = E[X(t)Y(t+\tau)];$ jointly WSS $\Rightarrow R_{XY}(\tau) = R_{YX}(-\tau)$                                                                                                                                                                                     |
| Gaussian process                             | every finite sample vector $[X(t_1) \cdots X(t_k)]'$ is Gaussian; WSS + Gaussian $\Rightarrow$ stationary                                                                                                                                                                        |

**The Professor's Three Flagged Chapter-13 Theorems, One More Time**

**Thm 13.10:** stationary  $X(t) \Rightarrow aX(t) + b$  stationary ( $a > 0$ ).      **Thm 13.11:** stationary  $\Rightarrow$  const. mean +  $R_X(t, \tau) = R_X(\tau)$ .      **Thm 13.12:** WSS  $\Rightarrow R_X(0) \geq 0$ , even, peak at 0.      **Ex. 13.22:**  $A \cos(2\pi f_c t + \Theta)$ ,  $\Theta \sim U(0, 2\pi)$ , is WSS with  $R_X(\tau) = \frac{A^2}{2} \cos 2\pi f_c \tau$ .

## 10 Rapid Self-Check Questions (Chapter 13, with answers)

1. A process where both  $X$  and  $t$  are continuous is a *continuous random process*.
2. A process whose future is fully determined by its past sample values is *deterministic*.
3. A process whose statistical properties don't change with time is *stationary*.
4. A first-order stationary process has a *constant mean*.
5.  $R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$ .
6. Ensemble averages equal time averages  $\Rightarrow$  *(mean-)ergodic*.
7. For WSS processes,  $R_{XX}(0) = E[X^2(t)]$ .
8.  $|R_{XY}(\tau)| \leq \sqrt{R_{XX}(0)R_{YY}(0)}$  **and**  $|R_{XY}(\tau)| \leq \frac{1}{2}[R_{XX}(0) + R_{YY}(0)]$  — both true (the first is tighter, since geometric mean  $\leq$  arithmetic mean).
9. A WSS noise process with constant power density spectrum is *white noise*.
10. Band-limited sampling frequency  $\omega_s = 2\pi/T_s$ .

## Final Reminders

- The professor's choice section draws from **Ch. 4, 5, 6** — know CDF/PDF basics, Gaussian RVs, delta/mixed RVs (Ch. 4); joint expectation, covariance/correlation, bivariate Gaussian (Ch. 5); and the  $F_W \rightarrow f_W$  derivation method including power/limiter problems (Ch. 6) cold.
- **Two guaranteed questions on Chapter 13** — prioritize Theorems 13.10, 13.11, 13.12, Example 13.22, the WSS definition, and the ergodicity concept above everything else in that chapter.
- Always ask three questions on a Ch. 13 problem: (1) Is the mean constant? (2) Does  $R_X$  depend on  $t$ , or only on  $\tau =$  the time difference? (3) If both yes, it's WSS — then check evenness/peak-at-zero (Thm 13.12) or ergodicity as asked.
- On Ch. 6 problems, always start with the two-step method: find  $F_W(w) = P[g(X) \leq w]$ , then differentiate — and watch for clipped/range-limited variables, which add an impulse at the clip point.